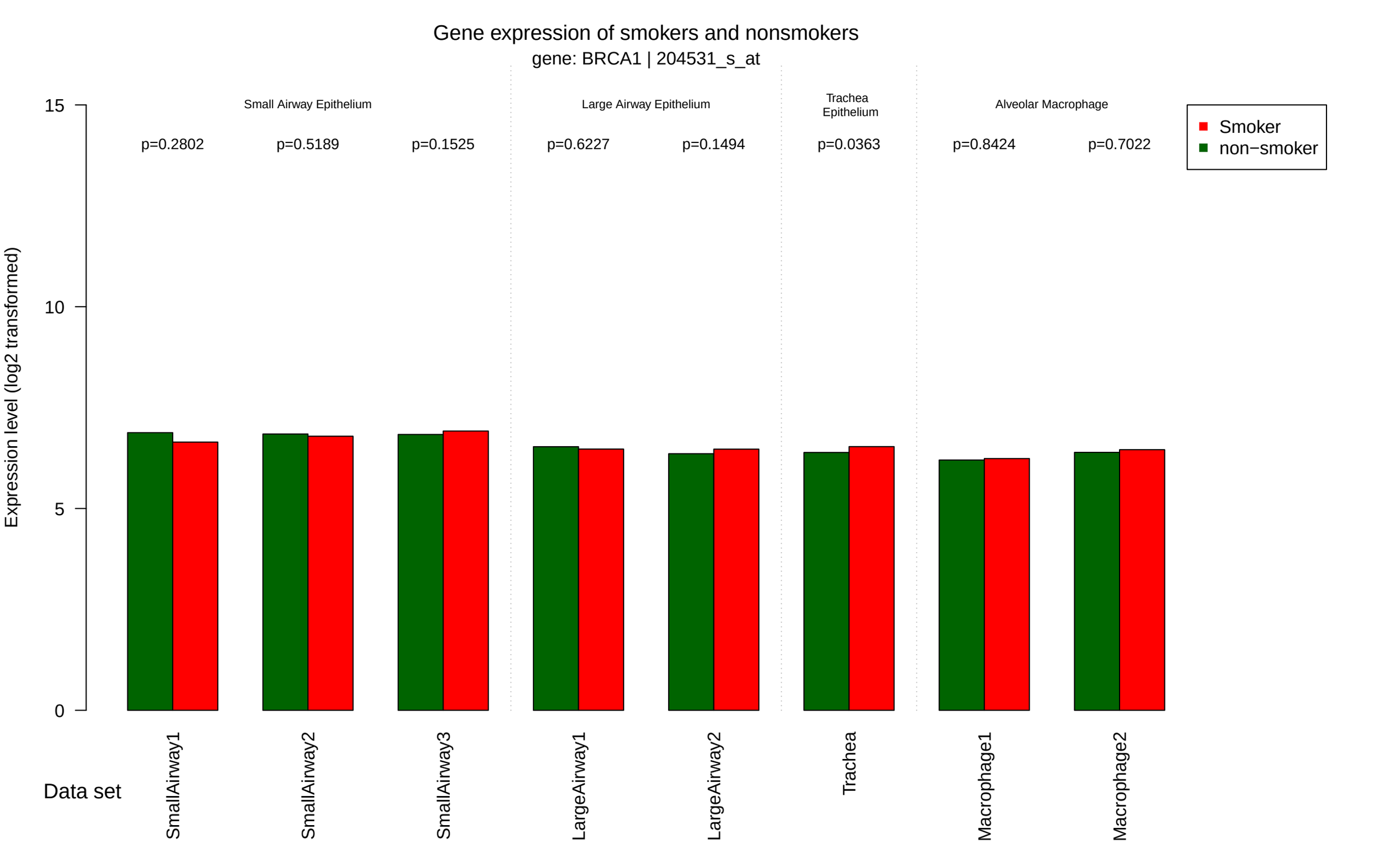
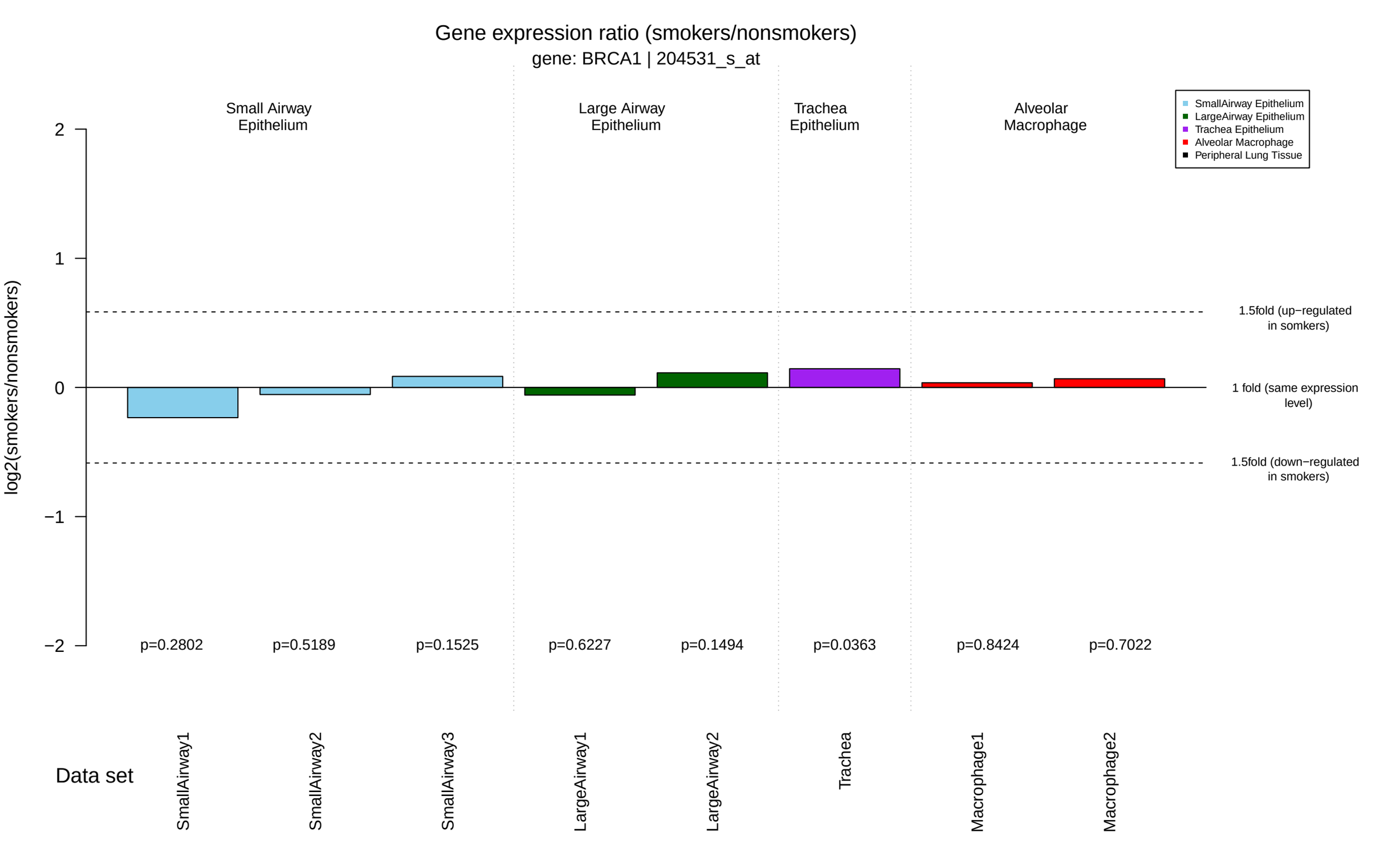
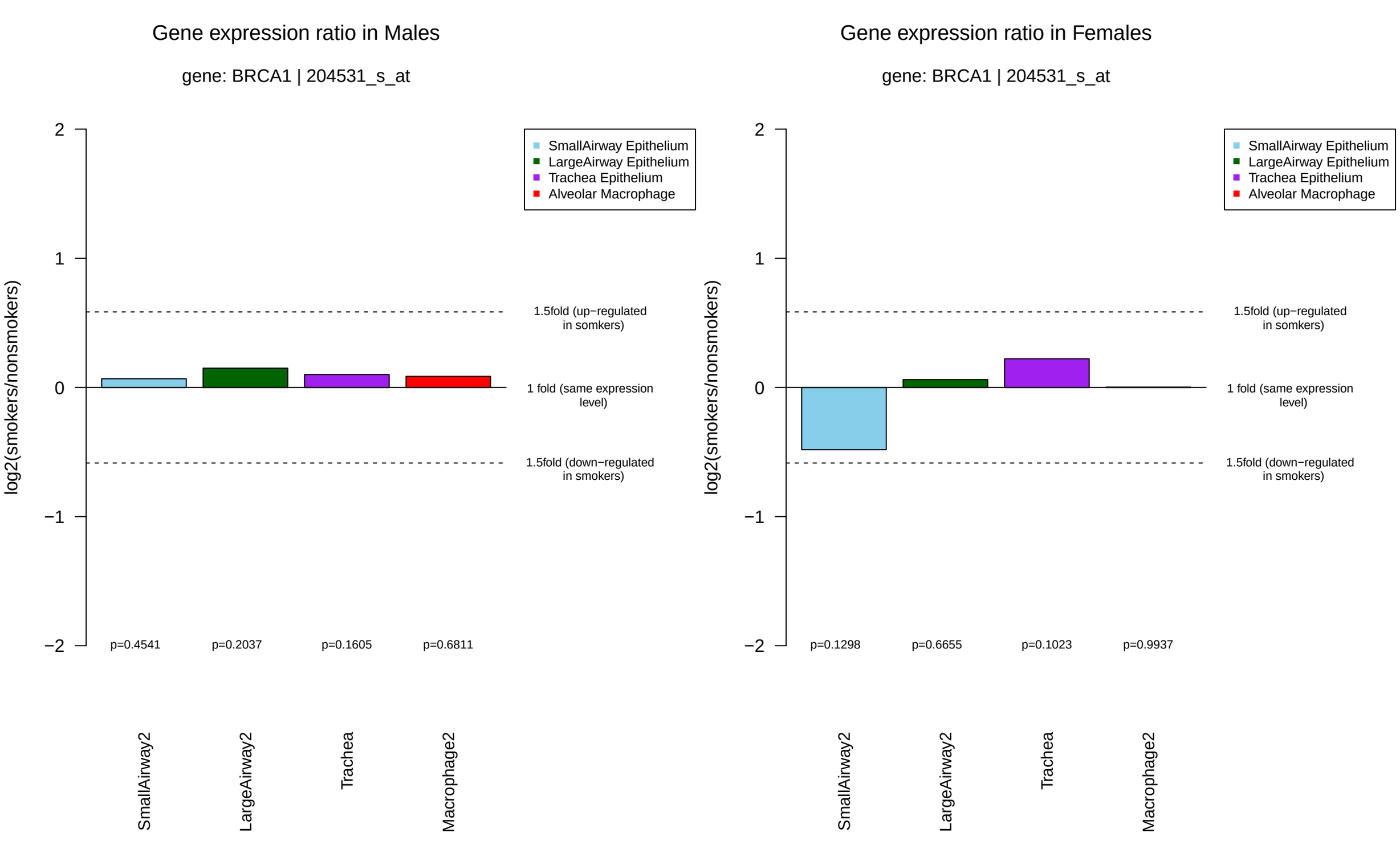
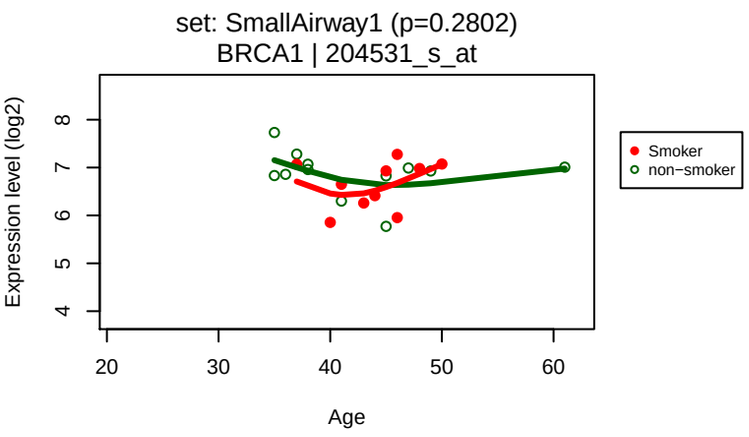
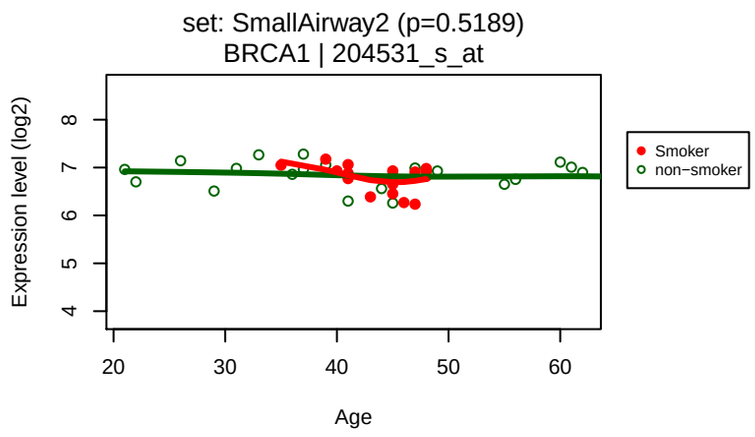
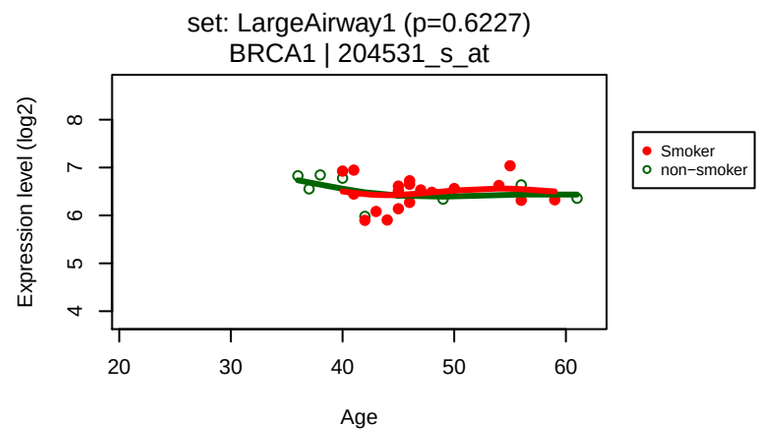

Gene expression of smokers and nonsmokers
gene: BRCA1 | 204531_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2802
p=0.5189
p=0.1525
p=0.6227
p=0.1494
p=0.0363
p=0.8424
p=0.7022
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

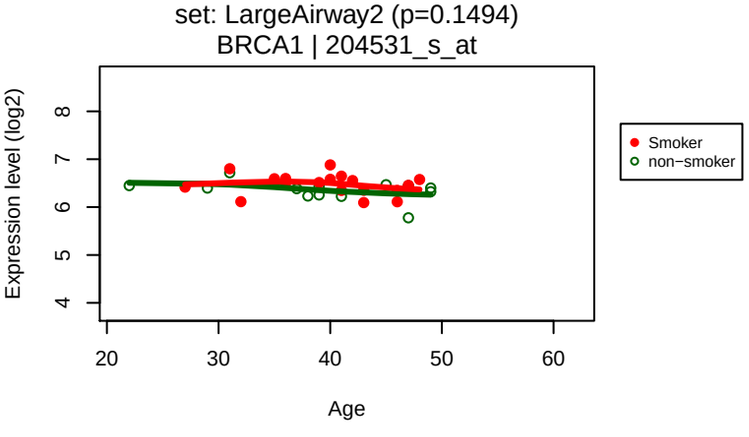

Gene expression ratio (smokers/nonsmokers)
gene: BRCA1 | 204531_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2802
p=0.5189
p=0.1525
p=0.6227
p=0.1494
p=0.0363
p=0.8424
p=0.7022
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

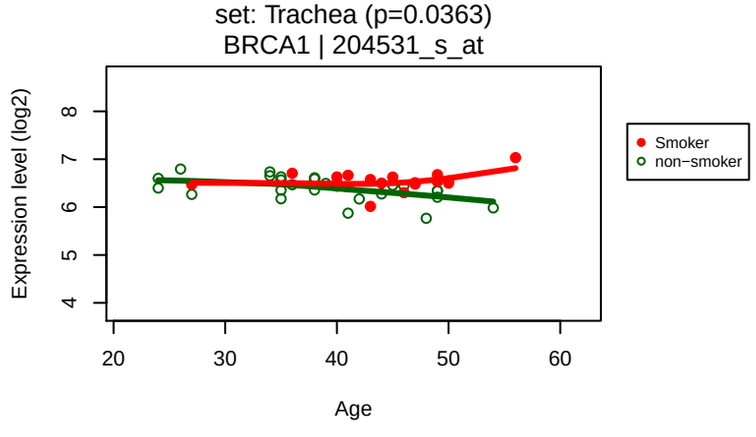

Gene expression ratio in Males
gene: BRCA1 | 204531_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4541
p=0.2037
p=0.1605
p=0.6811
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRCA1 | 204531_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1298
p=0.6655
p=0.1023
p=0.9937
SmallAirway2
LargeAirway2
Trachea
Macrophage2

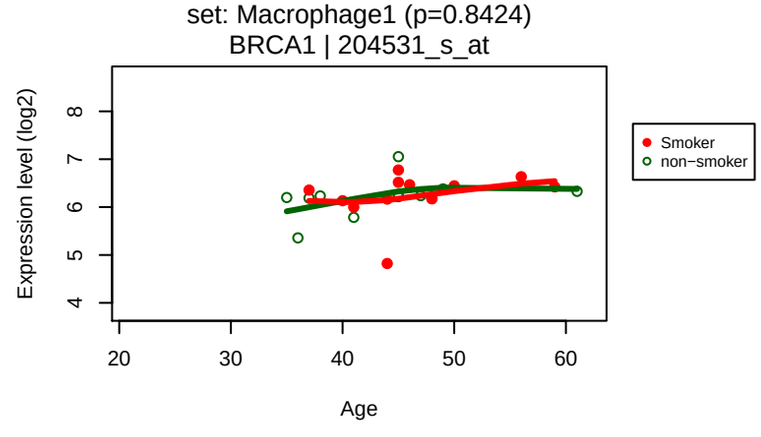

set: SmallAirway1 (p=0.2802)
BRCA1 | 204531_s_at
Expression level (log2)
Age
Smoker
non−smoker

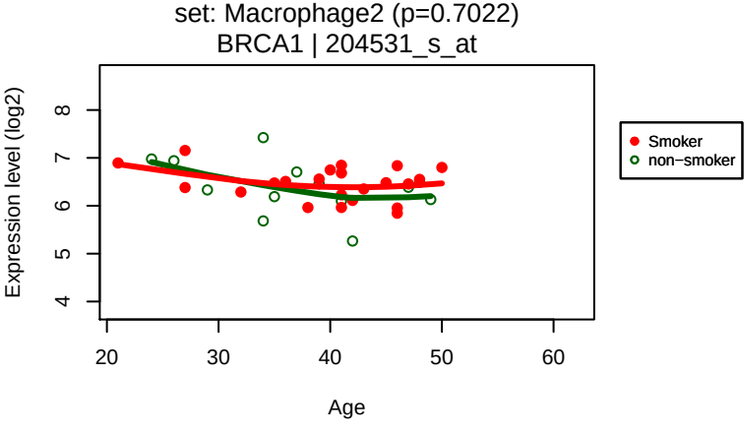

set: SmallAirway2 (p=0.5189)
BRCA1 | 204531_s_at
Expression level (log2)
Age
Smoker
non−smoker

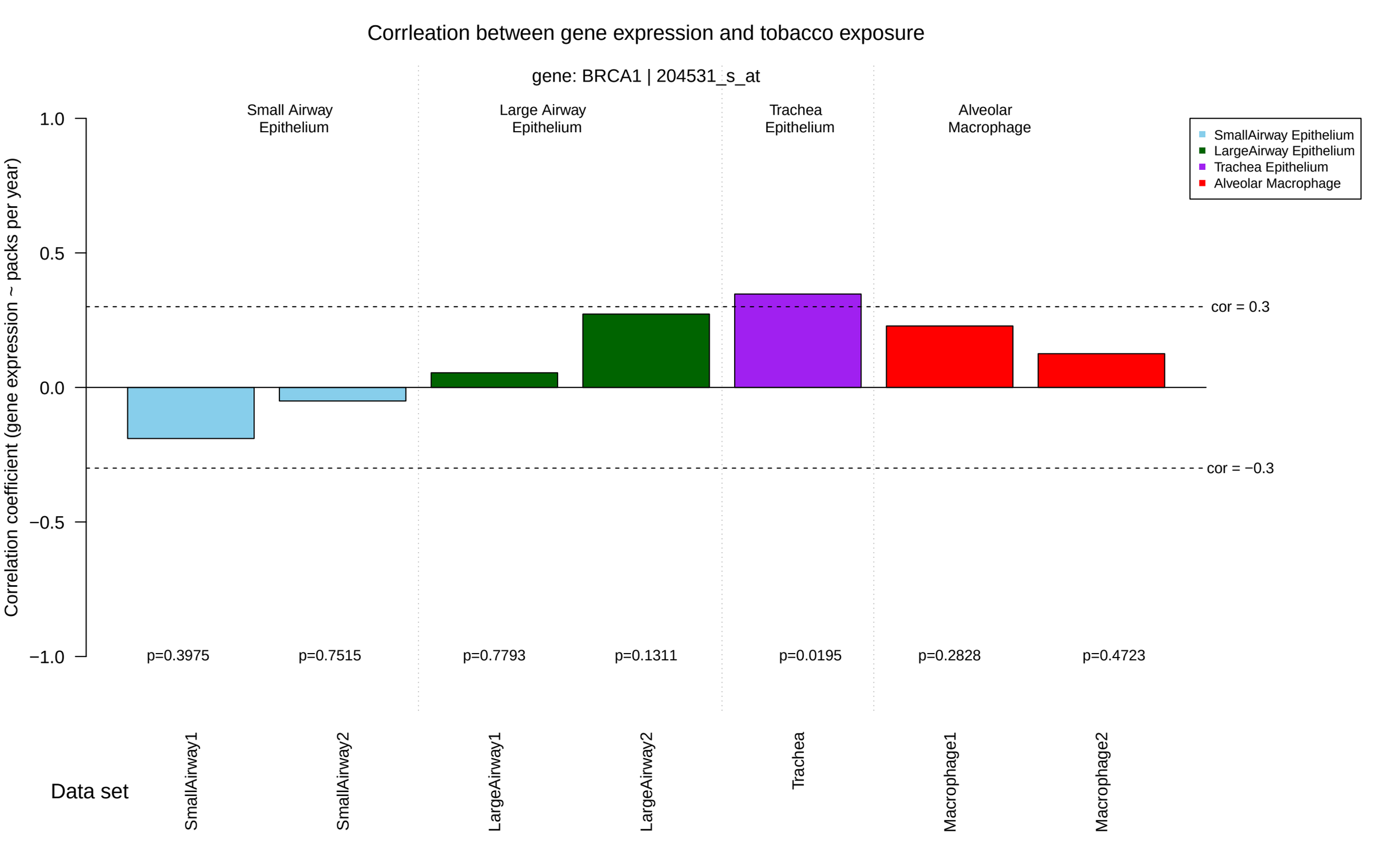

set: LargeAirway1 (p=0.6227)
BRCA1 | 204531_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1494)
BRCA1 | 204531_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0363)
BRCA1 | 204531_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8424)
BRCA1 | 204531_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7022)
BRCA1 | 204531_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRCA1 | 204531_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3975
p=0.7515
p=0.7793
p=0.1311
p=0.0195
p=0.2828
p=0.4723
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2